AQ8.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Consider two vectors $a = (1, -2, 2), b = (4, 0, -3)$a = (1, −2, 2), b = (4, 0, −3). Using the standard dot product as the inner product, the projection of aa in the direction of bb is given by

- $(\frac{-8}{25}, 0, \frac{6}{25})(25-8, 0, 256)$
- $(\frac{-8}{25}, \frac{6}{25})(25-8, 256)$
- $(\frac{8}{25}, 0, \frac{-6}{25})(258, 0, 25-6)$
- $(\frac{8}{25}, \frac{-6}{25})(258, 25-6)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{-8}{25}, 0, \frac{6}{25})(25-8, 0, 256)$

1 point

Consider an orthonormal basis $\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\}\{\quad 5$

√

$1(1, 2),\quad 5$

√

$1(-2, 1)\}$ of R^2 R2. If $(3, 5) = c_1 v_1 + c_2 v_2(3, 5)$ = c1v1 + c2v2 where $c_1, c_2 \in R$c1, c2 ∈ R and $v_1 = \frac{1}{\sqrt{5}}(1, 2), v_2 = \frac{1}{\sqrt{5}}(-2, 1)$v1 = $\quad 5$

√

$1(1, 2),$ v2 = $\quad 5$

√

$1(-2, 1)$, then choose the correct option.

- $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$c1 = $\quad 5$

√

$13,$ c2 = $\quad 5$

√

1

- $c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$c1 = $\quad 5$

√

$-13,$ c2 = $\quad 5$

√

1

- $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = $\quad 5$

√
13, c2 = 5

√
−1
○
$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5

√
−13, c2 = 5

√
−1

No, the answer is incorrect.
Score: 0
Accepted Answers:
$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5

√
13, c2 = 5

√
−1

1 point

Consider an orthonormal basis $\{(1,0,0),(0,1,0)\}\{(1,0,0),(0,1,0)\}$ for a subspace WW in R^3 R3. If $x = (1,2,3)$x = (1, 2, 3) is a vector in R^3 R3, then which of the following represents a vector in WW whose distance from xx is the least? Consider dot product as the standard inner product.

○
(2, 4, 0)(2, 4, 0)

○
(3, 4, 0)(3, 4, 0)

○
(4, 5, 0)(4, 5, 0)

○
(1, 2, 0)(1, 2, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1, 2, 0)(1, 2, 0)

Level 2

Consider the inner product $\langle a, b\rangle = a_1 b_1 - a_1 b_2 - a_2 b_1 + 4a_2 b_2$ ⟨a, b⟩ = a1b1 − a1b2 − a2b1 + 4a2 b2 where $a = (a_1, a_2), b = (b_1, b_2)$a = (a1, a2), b = (b1, b2) are vectors in R^2 R2. Use this information to answer questions 4 and 5.

1 point

Let $x = (1,2)$x = (1, 2). Find the projection of xx in the direction of $(3,4)$(3, 4) using the inner product defined above.

○
$\frac{25}{49}$(3, 4)4925(3, 4)

○
$\frac{11}{25}$(3, 4)2511(3, 4)

◯
$\frac{25}{49}(1, 2)$4925(1, 2)
◯
$\frac{11}{25}(1, 2)$2511(1, 2)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{25}{49}(3, 4)$4925(3, 4)

1 point
Let x = (1, 2)x = (1, 2), find the projection of xx in a direction perpendicular to (3, 4)(3, 4).

◯
$\frac{25}{49}(- 4, 3)$4925(−4, 3)
◯
$\frac{11}{25}(4, \ - 3)$2511(4, −3)
◯
$(- \frac{26}{49}, \ - \frac{2}{49})$(−4926, −492)
◯
$(\frac{26}{49}, \ - \frac{2}{49})$(4926, −492)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(- \frac{26}{49}, \ - \frac{2}{49})$(−4926, −492)

1 point
Consider an orthogonal basis {(1, 2, 1), (− 2, 0, 2)}{(1, 2, 1), (−2, 0, 2)} of a subspace WW, of the inner product space R^3 R3 with respect to the dot product. If y = (1, 2, 3) ∈ R^3 y = (1, 2, 3) ∈ R3, then find $Proj_W(y)$P rojW (y).

◯
$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$(31, 38, 37)
◯
$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})($ 6
√
18 − 8
√
32, 8
√
36, 6
√
18 + 8
√
32)
◯
$(\frac{1}{3}, \ - \frac{8}{3}, \frac{7}{3})$(31, −38, 37)
◯
$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \ - \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})($ 6
√

$18 - \sqrt{8}$

$\sqrt{32}, \sqrt{8}$

$\sqrt{36}, -\sqrt{6}$

$\sqrt{18} + \sqrt{8}$

$\sqrt{32}$)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})(31, 38, 37)$

Check Answers

Your score is: 0/6